

EyeVision AI — Refractive-error regressor evaluation

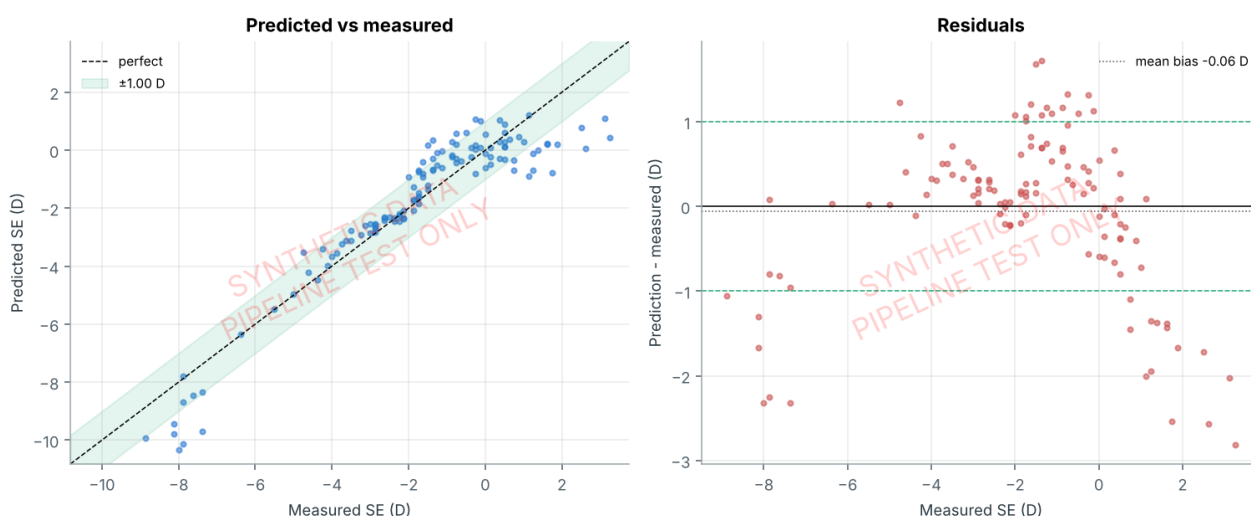
Model **ref-efficientnet_b0-synthetic-20261006-2158** (efficientnet_b0), trained 2026-10-06T22:08:13 on **synthetic**. Evaluated on: held-out TEST split (60 patients, 120 images; split by patient).

SYNTHETIC DATA — PIPELINE TEST ONLY. This model was trained and tested on artificially generated images. The numbers below verify that the code runs; they say nothing about real-world or clinical performance.

Research/educational AI estimate only. This system is not a medical diagnostic tool and does not provide a clinical diagnosis or eyeglass prescription. Please consult a qualified ophthalmologist/optometrist for diagnosis and refraction.

Spherical-equivalent regression

n	MAE (D)	MAE 95% CI	RMSE (D)	MSE	R ²	Bias (D)	±0.50 D	±1.00 D	±2.00 D
120	0.71	[0.59, 0.84]	0.96	0.92	0.863	-0.06	49.2%	70.8%	93.3%



An estimated SE is not an eyeglass prescription; subjective refraction by a professional is required.

Comparison with simple baselines (same test patients)

Model	MAE (D)	RMSE (D)	R ²	±1.00 D
efficientnet_b0 (this model)	0.71	0.96	0.863	70.8%
Mean-SE baseline	1.97	2.60	-0.001	33.3%
Linear (ridge) on image features + age	0.72	0.96	0.863	73.3%

Subgroup analysis (rows marked * are small; differences may be noise)

variable	group	n_images	n_patients	mae	rmse	bias	within_0.50D	within_1.00D
Age group	40-59	36	18	0.682	0.907	0.119	55.6%	69.4%
Age group	60+	52	26	0.886	1.146	-0.260	36.5%	63.5%
Age group	<40	32	16	0.461	0.636	0.067	62.5%	84.4%
Sex	F	64	32	0.759	1.021	-0.026	48.4%	64.1%
Sex	M	56	28	0.657	0.887	-0.097	50.0%	78.6%
Eye	L	60	60	0.777	1.033	-0.098	46.7%	66.7%
Eye	R	60	60	0.645	0.883	-0.021	51.7%	75.0%
Source dataset	synthetic	120	60	0.711	0.961	-0.059	49.2%	70.8%
Refraction band	high myopia (<= -6) *	11	6	1.238	1.469	-1.219	18.2%	45.5%
Refraction band	moderate (-6..-3) *	17	11	0.409	0.509	0.397	64.7%	94.1%

variable	group	n_images	n_patients	mae	rmse	bias	within_0.50D	within_1.00D
Refraction band	low myopia (-3..-0.5)	51	29	0.538	0.703	0.497	56.9%	76.5%
Refraction band	emmetropia (± 0.5) *	23	15	0.456	0.554	0.038	60.9%	91.3%
Refraction band	hyperopia ($> +0.5$) *	18	11	1.491	1.673	-1.481	16.7%	22.2%